

Bhavesh Ghanchi

Software Engineer

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EDUCATION

University of Southern California, Los Angeles, CA | MS Computer Science | GPA: 3.83/4.00 May 2026
K. J. Somaiya College of Engineering, Mumbai, India | BTech Computer Engineering | CGPA: 9.64/10 Jun 2024

EXPERIENCE

Easley-Dunn Productions

Torrance, CA (Remote)

Software Engineer Intern | Co-Lead, Multiplayer Core Gameplay Team

Jul 2026 - Present

- Co-lead the multiplayer core gameplay team for Robot Race, owning the networking and persistence layer across Firebase, Unity Gaming Services, Mirror, and EOS.
- Engineered event-driven C# integrations and RPC state propagation with fault-tolerant retries, keeping multiplayer sessions synchronized and preserving player state through network outages.

Streets Are For Everyone (SAFE)

Los Angeles, CA (Remote)

Software Developer Intern

Jun 2025 - Aug 2025

- Owned the organization's mapping platform end to end, replacing its static map with an interactive React and Node.js application.
- Built the Python geospatial ETL pipeline behind it, converting 497 raw bikeway features into 1,248 routable segments: a 151% gain in routing granularity at 100% route coverage.
- Shipped an LLM-powered route recommendation feature to a 60-cyclist beta, translating natural-language requests into safer routes from the classified GIS layer, and iterated on rider feedback before wider rollout.

Colgate-Palmolive

Mumbai, India

Full-Stack Intern

Jan 2024 - Jun 2024

- Built a 25+ component React/TypeScript design system that the rebuilt Hills Pet Hub site was assembled from, standardizing UI across markets and cutting frontend development time.
- Launched the site in France to 10,000+ users on day one, with rollout to 6 further countries scheduled over the next 3 months on the same component library.
- Optimized SQL queries and added Redis caching behind the site's REST APIs, cutting peak query latency by 25%.

K. J. Somaiya College of Engineering

Mumbai, India

Research Intern

Mar 2023 - May 2023

- Built a Python pipeline to scrape and structure text and image data from PDF, HTML, and Word documents, applying NLTK and SciSpacy clinical NER for healthcare domain classification.

SayHey

Mumbai, India

Backend Developer Intern

Jun 2022 - Aug 2022

- Built and optimized booking microservice workflows for a mental health app, cutting latency by 30% and reducing costs ~15% by replacing third-party scheduling software.

PROJECTS

OTAS: Observability Tool for Agentic AI Systems | github.com/bhaveshGhanchi/otas

Apr 2026

Python, Django, PostgreSQL, Redis, Celery, OpenTelemetry, MCP, Docker | Team project + personal fork

- Co-built an observability platform for AI agents capturing prompts, tool calls, latency, and outcomes, linked into execution lineage across runs, handoffs, and retries.
- Added OpenTelemetry tracing, a Pydantic-validated MCP server, and CLI hooks for Claude Code, Cursor, and Codex; load-tested at 10 agents and 800 sessions, persisting 1,620 events at 149 ms p95.

LEAP: Loss-Aware End-to-End Acknowledged Protocol | github.com/bhaveshGhanchi/leap

May 2026

Java, UDP Sockets, Maven

- Implemented TCP-style reliable transport from scratch over UDP: sliding window, cumulative ACKs, fast retransmit, adaptive RTO, Tahoe congestion control, and CRC32 plus end-to-end SHA-256 integrity checks.
- Built a benchmarking harness with three packet-loss simulation modes, measuring 46.2 MB/s at 0% loss and 88.8% efficiency at 10% loss; documented the design in a 6-part blog series.

PUBLICATIONS

M. Garg, L. Wang, **B. Ghanchi**, S. Dumpala, S. Kakde, Y. Chen. "Biomedical Literature Q&A System Using RAG." *arXiv preprint arXiv:2509.05505*, Sep 2025.

- RAG medical Q&A pipeline: MiniLM embeddings, top-5 FAISS retrieval, Mistral-7B-v0.3 fine-tuned with QLoRA on MedQuAD; BERTScore-F1 0.88-0.90 with a breast-cancer-only index.

SKILLS

Languages: Python, Java, C#, TypeScript, JavaScript, SQL

AI & ML: RAG, Embeddings, FAISS Vector Search, QLoRA Fine-Tuning, Hugging Face, LLM Evals, Agentic Workflows, OpenAI API, MCP

Backend & Data: FastAPI, Django, Node.js, Express, React, REST APIs, Event-Driven Microservices, PostgreSQL, Redis, Celery, Firebase

Infrastructure: Docker, Git, CI/CD, Linux, OpenTelemetry, Claude Code, Cursor